

Python Basics

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Objective: Learn Python fundamentals for Data Science - Part 1

```
In [31]: # =====  
# STEP 1: Hello World  
# =====  
print("Hello, Python!")  
print(10 + 5)
```

Hello, Python!
15

```
In [3]: #=====  
# STEP 2: Variables  
# =====  
a = 10  
b = 20  
print("Value of a:", a)  
print("Value of b:", b)
```

Value of a: 10
Value of b: 20

```
In [5]: name = "KL University"  
print("Name:", name)
```

Name: KL University

```
In [48]: # =====  
# STEP 3: Data Types  
# =====  
x = 10  
y = 10.5  
text = "Python"  
flag = True  
  
print(type(x))  
print(type(y))  
print(type(text))  
print(type(flag))
```

```
<class 'int'>
<class 'float'>
<class 'str'>
<class 'bool'>
```

```
In [12]: # =====
# STEP 4: User Input
# =====
# Uncomment to test
num = int(input("Enter a number: "))
print("You entered:", num)
```

Enter a number: 2
You entered: 2

```
In [13]: # =====
# STEP 5: Conditional Statements
# =====
num = 15

if num > 10:
    print("Greater than 10")
else:
    print("Less than or equal to 10")
```

Greater than 10

```
In [14]: # =====
# STEP 6: Loops
# =====
print("For Loop:")
for i in range(5):
    print(i)
```

For Loop:
0
1
2
3
4

```
In [15]: print("While Loop:")
i = 1
while i <= 5:
    print(i)
    i += 1
```

While Loop:
1
2
3
4
5

```
In [16]: # =====
# STEP 7: Lists
# =====
numbers = [10, 20, 30, 40]
print(numbers[0])
```

10

```
In [17]: numbers.append(50)
```

```
In [18]: for num in numbers:
        print(num)
```

```
10
20
30
40
50
```

```
In [22]: # =====
# STEP 8: Dictionaries
# =====
student = {
    "name": "Hanish",
    "age": 21,
    "marks": 90
}
```

```
In [23]: print(student["name"])

student["grade"] = "A"
print(student)
```

```
Hanish
{'name': 'Hanish', 'age': 21, 'marks': 90, 'grade': 'A'}
```

```
In [24]: # =====
# STEP 9: Functions
# =====
def add(a, b):
    return a + b
```

```
In [25]: result = add(10, 20)
        print("Sum:", result)
```

```
Sum: 30
```

```
In [26]: # =====
# STEP 10: Programs
# =====
# Even or Odd
num = 7
if num % 2 == 0:
    print("Even")
else:
    print("Odd")
```

```
Odd
```

```
In [27]: # Sum of numbers
        total = 0
        for i in range(1, 6):
            total += i
        print("Sum:", total)
```

```
Sum: 15
```

```
In [28]: # Largest number
        a = 10
        b = 25
        c = 15

        if a > b and a > c:
            print("A is largest")
```

```
elif b > c:
    print("B is largest")
else:
    print("C is largest")
```

B is largest

In [30]: `print("Lab Completed Successfully!")`

Lab Completed Successfully!

=====

NUMPY LAB (Python for Data Science - Part 2)

=====



In [32]: `#STEP 1: Import NumPy`
`import numpy as np`

In [33]: `# STEP 2: Create Arrays`
`arr1 = np.array([1, 2, 3, 4, 5])`
`print("Array:", arr1)`

Array: [1 2 3 4 5]

In [35]: `# 2D Array`
`arr2 = np.array([[1, 2, 3], [4, 5, 6]])`
`print("2D Array:", arr2)`

2D Array: [[1 2 3]
[4 5 6]]

In [36]: `# STEP 3: Array Properties`
`print("Shape:", arr2.shape)`
`print("Dimensions:", arr2.ndim)`
`print("Data Type:", arr2.dtype)`

Shape: (2, 3)
Dimensions: 2
Data Type: int32

In [38]: `# STEP 4: Special Arrays`
`zeros = np.zeros((2,3))`
`ones = np.ones((2,3))`
`identity = np.eye(3)`

`print("Zeros:", zeros)`
`print("Ones:", ones)`
`print("Identity Matrix:", identity)`

```
Zeros: [[0. 0. 0.]
        [0. 0. 0.]]
Ones: [[1. 1. 1.]
        [1. 1. 1.]]
Identity Matrix: [[1. 0. 0.]
                  [0. 1. 0.]
                  [0. 0. 1.]]
```

```
In [39]: # STEP 5: Array Operations
arr = np.array([10, 20, 30, 40])
print("Addition:", arr + 5)
print("Multiplication:", arr * 2)
print("Square:", arr ** 2)
```

```
Addition: [15 25 35 45]
Multiplication: [20 40 60 80]
Square: [ 100  400  900 1600]
```

```
In [40]: # STEP 6: Mathematical Functions
print("Mean:", np.mean(arr))
print("Median:", np.median(arr))
print("Standard Deviation:", np.std(arr))
```

```
Mean: 25.0
Median: 25.0
Standard Deviation: 11.180339887498949
```

```
In [41]: # STEP 7: Indexing & Slicing
print("First element:", arr[0])
print("Slice:", arr[1:3])
```

```
First element: 10
Slice: [20 30]
```

```
In [43]: # STEP 8: Reshaping
arr3 = np.arange(1, 10)
reshaped = arr3.reshape(3,3)
print("Array", arr3)
print("Reshaped:", reshaped)
```

```
Array [1 2 3 4 5 6 7 8 9]
Reshaped: [[1 2 3]
           [4 5 6]
           [7 8 9]]
```

```
In [44]: # STEP 9: Random Numbers
rand = np.random.rand(3,3)
print("Random:", rand)
```

```
Random: [[0.42714771 0.26519545 0.92042356]
         [0.70625982 0.19858691 0.23996524]
         [0.61340165 0.07679773 0.77338076]]
```

```
In [46]: # STEP 10: Aggregation
print("Array", arr)
print("Sum:", np.sum(arr))
print("Max:", np.max(arr))
print("Min:", np.min(arr))
```

```
Array [10 20 30 40]
Sum: 100
Max: 40
Min: 10
```

```
In [47]: print("NumPy Lab Completed!")
```

=====

PANDAS LAB (Python for Data Science - Part 3)

=====



```
In [54]: # STEP 1: Import Pandas
import pandas as pd
```

```
In [50]: # STEP 2: Create DataFrame
# From dictionary
student_data = {
    'Name': ['A', 'B', 'C', 'D'],
    'Age': [20, 21, 19, 22],
    'Marks': [85, 90, 78, 92]
}
```

```
In [51]: student_data
```

```
Out[51]: {'Name': ['A', 'B', 'C', 'D'],
          'Age': [20, 21, 19, 22],
          'Marks': [85, 90, 78, 92]}
```

```
In [62]: df = pd.DataFrame(student_data)
```

```
In [64]: df
```

```
Out[64]:
```

	Name	Age	Marks
0	A	20	85
1	B	21	90
2	C	19	78
3	D	22	92

```
In [66]: # STEP 3: Basic Inspection
df.head()
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 3 columns):
#   Column  Non-Null Count  Dtype
---  -
0   Name    4 non-null           object
1   Age     4 non-null           int64
2   Marks   4 non-null           int64
dtypes: int64(2), object(1)
memory usage: 228.0+ bytes
```

Out[66]:

	Age	Marks
count	4.000000	4.000000
mean	20.500000	86.250000
std	1.290994	6.238322
min	19.000000	78.000000
25%	19.750000	83.250000
50%	20.500000	87.500000
75%	21.250000	90.500000
max	22.000000	92.000000

In [67]: `df['Marks']`

Out[67]:

0	85
1	90
2	78
3	92

Name: Marks, dtype: int64

In [69]: `df[['Name', 'Marks']]`

Out[69]:

	Name	Marks
0	A	85
1	B	90
2	C	78
3	D	92

In [71]: `#df.iloc[row_index, column_index]`
`df.iloc[0] # selects First row`

Out[71]:

Name	A
Age	20
Marks	85

Name: 0, dtype: object

In [73]: `df.iloc[0:3] #Select Multiple Rows`

Out[73]:

	Name	Age	Marks
0	A	20	85
1	B	21	90
2	C	19	78

In [75]: `df.iloc[0,1] # intestsection element of first and second rows`

Out[75]: 20

In [77]: `df.iloc[:,1] # selects entire second column`

```
Out[77]: 0    20
         1    21
         2    19
         3    22
         Name: Age, dtype: int64
```

```
In [79]: df.iloc[:,0:2] #Select Entire Multiple Columns
```

```
Out[79]:
```

	Name	Age
0	A	20
1	B	21
2	C	19
3	D	22

```
In [81]: df
```

```
Out[81]:
```

	Name	Age	Marks
0	A	20	85
1	B	21	90
2	C	19	78
3	D	22	92

```
In [86]: df.iloc[0:2,0:2] # intersection elements of Rows 1st ,2nd and columns 2nd ,3
```

```
Out[86]:
```

	Name	Age
0	A	20
1	B	21

```
In [87]: df.iloc[-1] # Select Last Row
```

```
Out[87]: Name      D
         Age      22
         Marks    92
         Name: 3, dtype: object
```

```
In [90]: # STEP 5: Add / Modify Columns
         df['Grade'] = ['B','A','C','A']
         df
```

```
Out[90]:
```

	Name	Age	Marks	Grade
0	A	20	85	B
1	B	21	90	A
2	C	19	78	C
3	D	22	92	A

```
In [91]: # Modify values
         df['Marks'] = df['Marks'] + 5
```


In [92]: df

Out[92]:

	Name	Age	Marks	Grade
0	A	20	90	B
1	B	21	95	A
2	C	19	83	C
3	D	22	97	A

0	A	20	90	B
1	B	21	95	A
2	C	19	83	C
3	D	22	97	A

In [95]: *# STEP 6: Filtering Data*
high_marks = df[df['Marks'] > 85]
high_marks

Out[95]:

	Name	Age	Marks	Grade
0	A	20	90	B
1	B	21	95	A
3	D	22	97	A

0	A	20	90	B
1	B	21	95	A
3	D	22	97	A

In [98]: *# Create sample with missing*
sample = pd.DataFrame({
 'A': [1, 2, None, 4],
 'B': [5, None, 7, 8]
})

sample.isnull().sum()

Out[98]: A 1
B 1
dtype: int64

In [101... *# Fill missing*
sample['A'] = sample['A'].fillna(sample['A'].mean())
sample['B'] = sample['B'].fillna(0)

sample

Out[101]:

	A	B
0	1.000000	5.0
1	2.000000	0.0
2	2.333333	7.0
3	4.000000	8.0

In [102... *# STEP 9: GroupBy Operation*
group = df.groupby('Grade')['Marks'].mean()
print("Average Marks by Grade:", group)

Average Marks by Grade: Grade
A 96.0
B 90.0
C 83.0
Name: Marks, dtype: float64

```
In [104... # STEP 10: Sorting
sorted_df = df.sort_values(by='Marks', ascending=False)
sorted_df
```

Out[104]:

Name	Age	Marks	Grade
------	-----	-------	-------

3	D	22	97	A
1	B	21	95	A
0	A	20	90	B
2	C	19	83	C

```
In [105]: print("Pandas Lab Completed!")
```

Pandas Lab Completed!

=====

DATA VISUALIZATION LAB (Part 4)

=====



```
In [106... import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [107... # Use seaborn dataset for visualization
viz_df = sns.load_dataset('titanic')
```

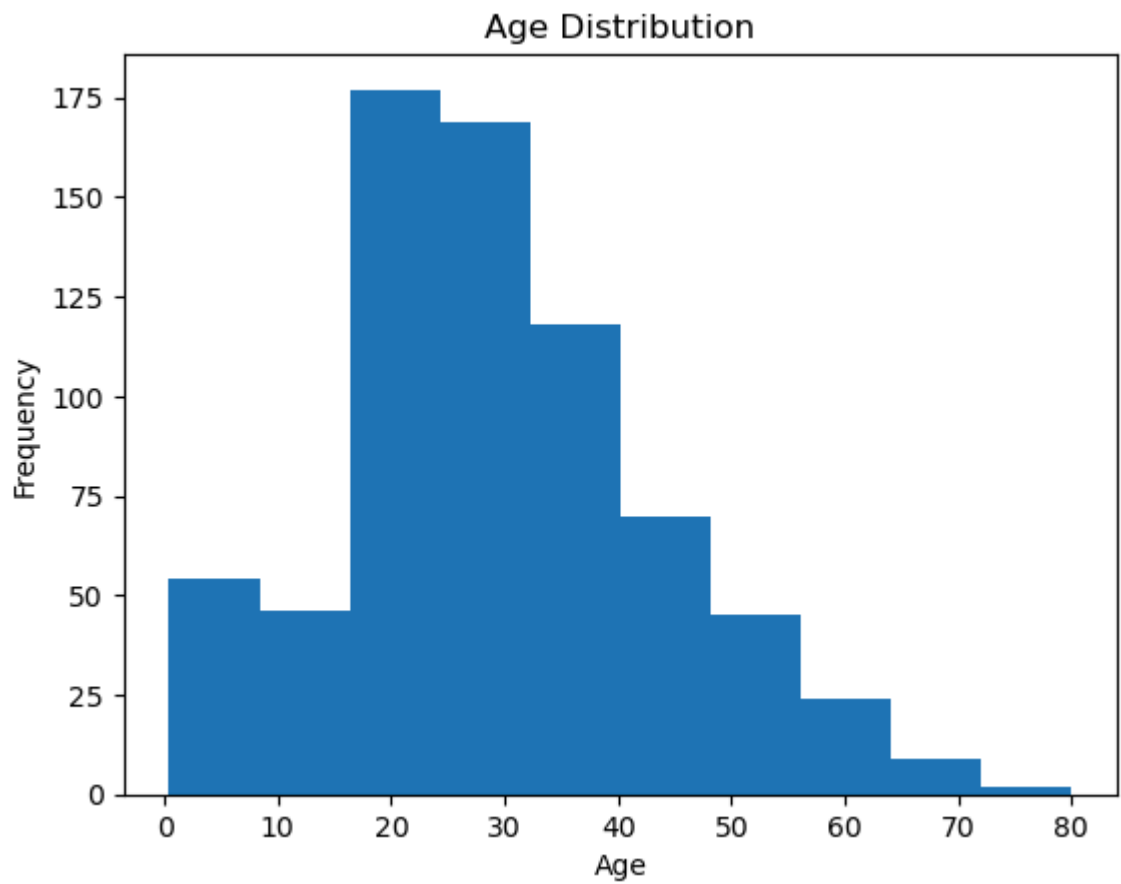
```
In [111... viz_df.head()
```

```
Out[111]:      survived  pclass      sex  age  sibsp  parch      fare  embarked  class      who  adult_male  de
```

0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	Na
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	Na
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	Na

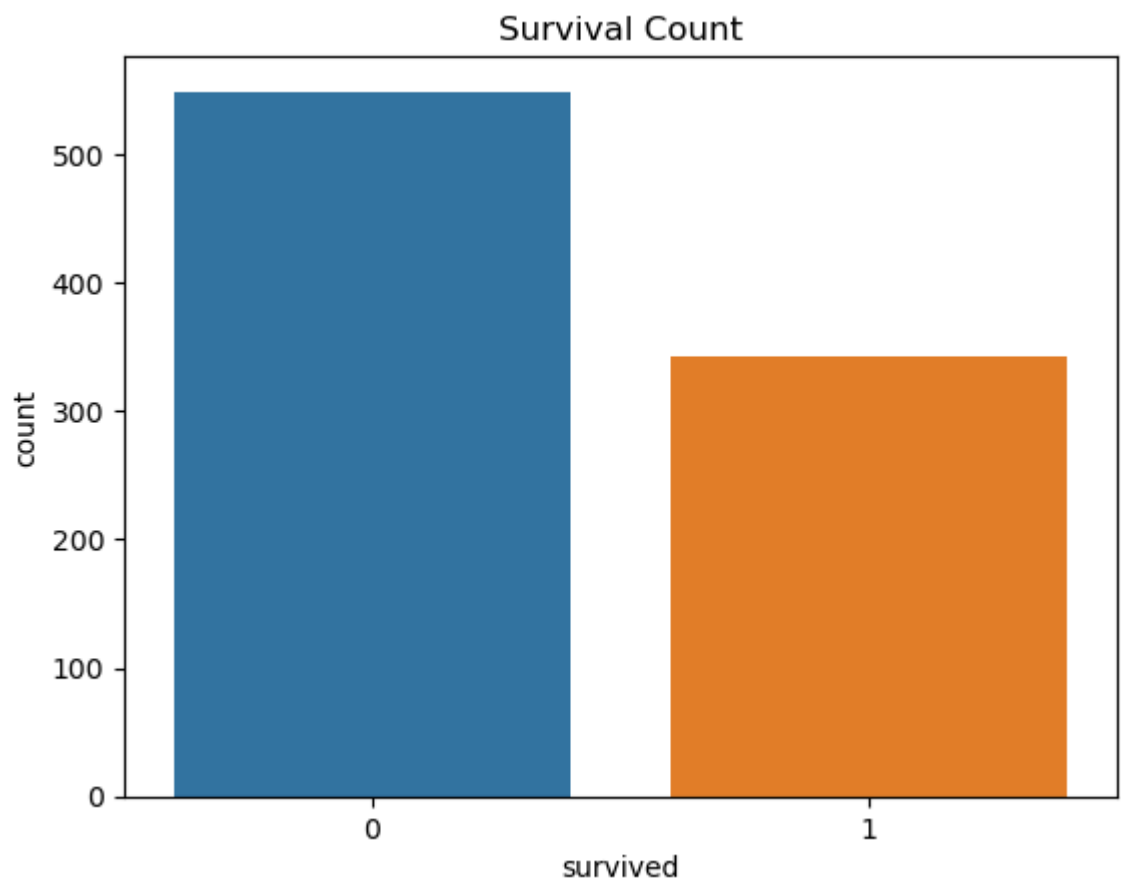


```
In [108... # STEP 2: Histogram
plt.figure()
plt.hist(viz_df['age'].dropna())
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```



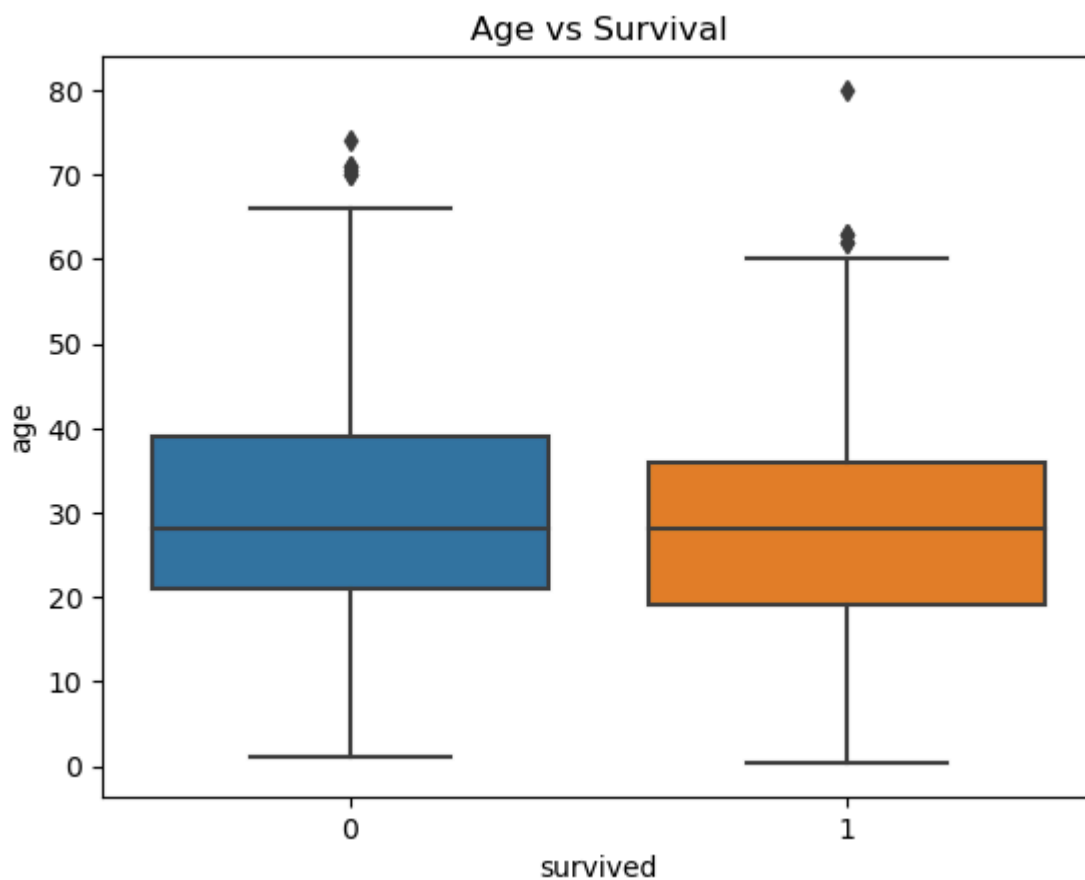
In [109...

```
# STEP 3: Count Plot  
plt.figure()  
sns.countplot(x='survived', data=viz_df)  
plt.title("Survival Count")  
plt.show()
```



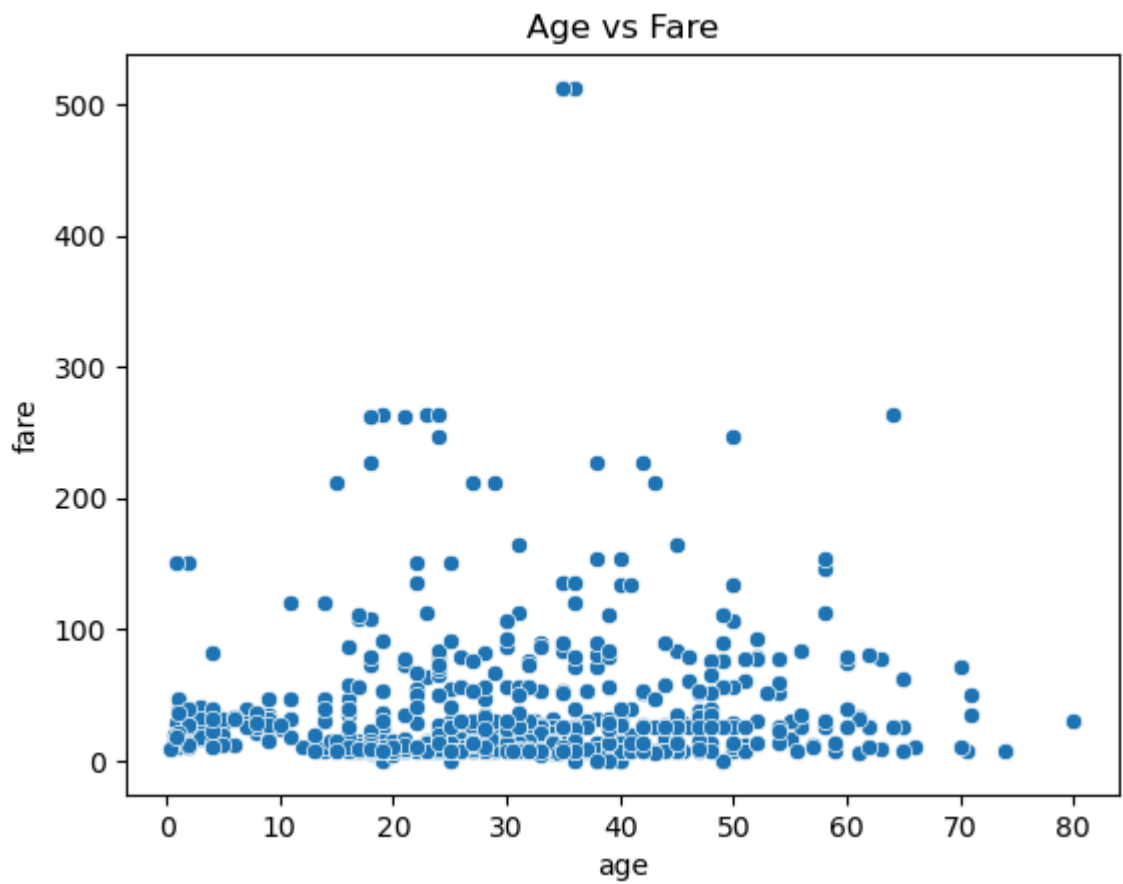
In [112...

```
# STEP 4: Box Plot
plt.figure()
sns.boxplot(x='survived', y='age', data=viz_df)
plt.title("Age vs Survival")
plt.show()
```



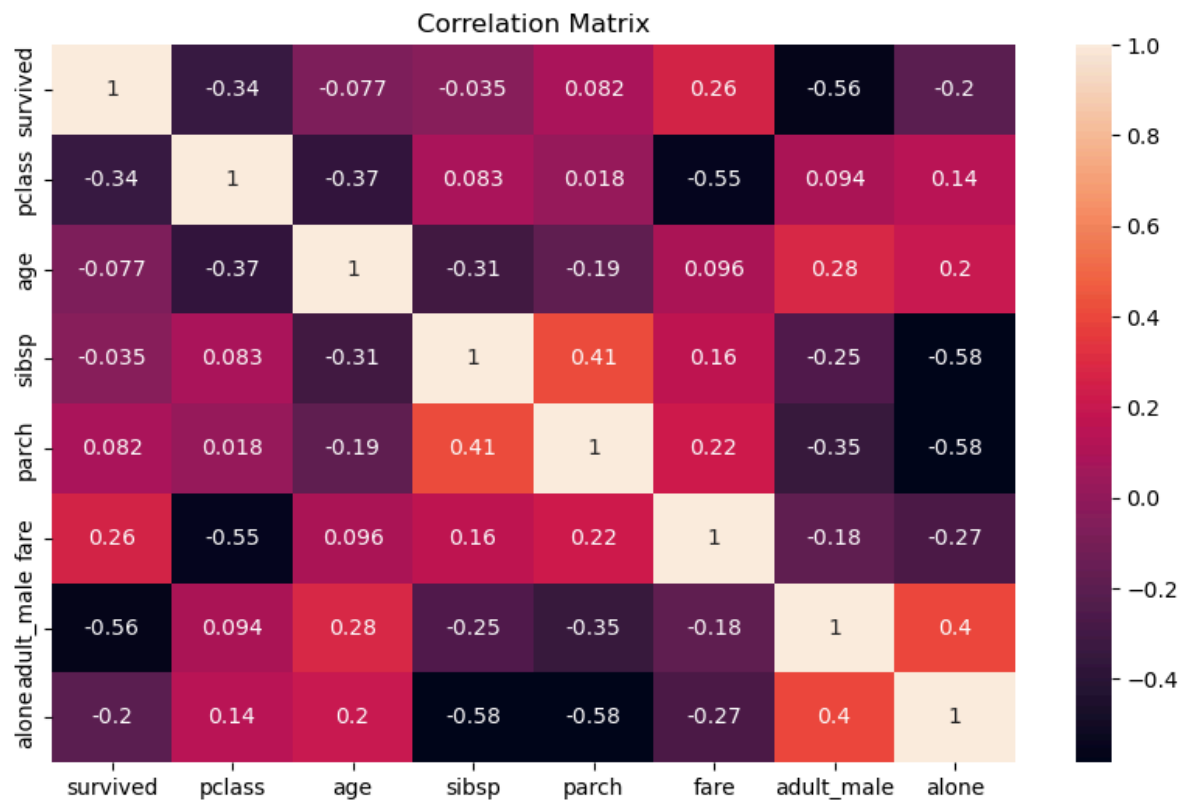
In [113...

```
# STEP 5: Scatter Plot
plt.figure()
sns.scatterplot(x='age', y='fare', data=viz_df)
plt.title("Age vs Fare")
plt.show()
```



In [114...]

```
# STEP 6: Correlation Heatmap
plt.figure(figsize=(10,6))
sns.heatmap(viz_df.corr(numeric_only=True), annot=True)
plt.title("Correlation Matrix")
plt.show()
```



In []: